Fourier Series

```
f(x) = a_0 / 2 + sum_{n=1}^{infty} [ a_n cos(n x) + b_n sin(n x) ]
a_n = (1 / pi) integral_{-pi}^{pi} f(x) cos(n x) dx
b_n = (1 / pi) integral_{-pi}^{pi} f(x) sin(n x) dx
Parseval: (1 / pi) integral |f|^2 = a_0^2 / 2 + sum (a_n^2 + b_n^2)
```

Convergence is pointwise where f is continuous, in L2 everywhere.

Matrix Form of Least Squares

```
minimise || A x - b ||_2^2  over x in R^n
normal equations:  A^T A x = A^T b
x_hat = (A^T A)^{-1} A^T b   when A has full column rank
projection matrix:  P = A (A^T A)^{-1} A^T,  P^2 = P = P^T
```

Escaped LaTeX in the source: \\frac{1}{2}, \\begin{bmatrix} 1 & 0 \\\\ 0 & 1 \\end{bmatrix}

Time Dilation

```
special relativity:   dt_SR = - v^2 / (2 c^2) per unit time
general relativity:   dt_GR = + G M / (r c^2) per unit time
net for a MEO satellite: +38 microseconds per day
positional drift:  c * 38e-6 s = 11.4 km per day
```

A 1 ns timing error is 30 cm of range error. Currency check: $1,200 per receiver.

Eigen-decomposition

```
A v = lambda v,   det(A - lambda I) = 0
A = Q Lambda Q^{-1}   for diagonalisable A
symmetric A: Q orthogonal, Lambda real
spectral radius rho(A) = max |lambda_i| governs stability
```

Worked example territory: pick a 2x2 and derive both eigenvalues.